



Abel Tesfaye Tsega

Passport: E00549350 **Nationality:** Ethiopian **Date of birth:** 28/06/2003

Gender: Male

 **Email:** abel.tesfaye@bdu.edu.et

 **LinkedIn:** [Abel Tesfaye](#)

 **Website:** <https://github.com/abel-tesfaye-geo>

 **Home:** (Ethiopia)

ABOUT ME

Gold Medalist and Assistant Lecturer at Bahir Dar University, with a BSc in Land Administration and Surveying and hands-on expertise in UAV photogrammetry, LiDAR processing, machine learning-based geospatial analysis, and cloud-based Earth observation. Built Python and JavaScript workflows for urban spatial expansion research using Google Earth Engine, and designed and built a functional drone prototype. Actively engaged in international professional development through NASA ARSET, ESA EO College, and ITC University of Twente. Seeking the Copernicus Master in Digital Earth, specializing in Geovisualisation and Geocommunication, to advance cartographic and web-based visualization expertise and help build Africa's geospatial capacity from within.

EDUCATION AND TRAINING

BSc in Land administration and surveying

Bahir Dar University [03/03/2021 – 22/06/2025]

City: Bahir Dar | **Country:** Ethiopia

BSc in Land Administration and Surveying.

Bahir Dar University — Institute of Land Administration <https://ila.bdu.edu.et>

- CGPA: 3.97/4.00 — Gold Medal, Top Graduate of Cohort
- National Exit Examination: 93/100 — Ranked 1st Nationally

WORK EXPERIENCE

 **Bahir Dar University** – Bahir Dar, Ethiopia

Assistant lecturer

[01/07/2025 – Current]

<https://www.bdu.edu.et/>

- Teach Photogrammetry and Remote Sensing to undergraduate students
- Design practical exercises in GEE, UAV data processing, and geospatial analysis
- Mentor students in research methodology and geospatial tool application
- Actively engaged in departmental academic development initiatives

 **Space Science and Geospatial Institute(SSGI)** <https://www.ssgi.gov.et/> – Addis Ababa, Ethiopia



Geospatial Intern

[01/06/2024 – 25/11/2024]

- Prepared flight mission planning for aerial surveying and executed simulated flight operations
- Processed raw UAV imagery into orthomosaics and DSMs using TrackAIR software
- Acquired and processed LiDAR point cloud data using Trimble Inpho
- Conducted geospatial work for the city of Bishoftu using geospatial software
- Gained hands-on experience in photogrammetry, LiDAR data acquisition and processing, and geospatial data quality control

 **Equator Geospatial work service P.L.C** – Bahir Dar, Ethiopia

GIS Analyst

[04/06/2023 – 08/12/2023]

- Generated 3D terrain and orthophoto visualizations from DJI Mavic 3E imagery for infrastructure planning and land development projects.
- Performed GNSS data processing and quality control for spatial datasets used in the development of the Bahir Dar City Structural Plan.

PROJECTS

Analysis of Urban Spatial Expansion Using GEE and Random Forest Machine Learning • Applied Google Earth Engine and Random Forest algorithm to model and quantify urban sprawl in Bahir Dar City and Presented findings at departmental research seminar

UAV Prototype for Cadastral Mapping

- Led the full design, engineering, and assembly of a quadcopter drone for land surveying applications

[10/07/2026 – Current]

UAV photogrammetry digital twin project at Geospatial Data and Technology Center(Bahir Dar University) Developing a UAV photogrammetry pipeline to capture aerial imagery and generate high-resolution 3D digital twins for accurate spatial analysis and visualization.

SKILLS

Geospatial and Remote Sensing

ArcGIS / QGIS / Agisoft metashape / Trimble Inpho / Google Earth Engine / AutoCAD / LiDAR Data Acquisition and Processing / Drone Piloting / UAV Flight Mission Planning / TrackAIR

Survey Instruments

Total Stations / GNSS RTK / Digital and Automatic Levels / Field Data Collection and Survey Documentation / CORS Networks

Programming

JavaScript / Python



Personal

Academic Mentorship / Research Leadership / Blender / Adobe Photoshop

LEADERSHIP AND COMMUNITY ROLES

[13/09/2023 – 16/05/2025]

President

Institute of Land Administration Student Association, BDU

- Led the student body, organized academic events, and fostered inter-departmental collaboration for two years

[02/09/2024 – 01/06/2025]

Head of Food Affairs and Welfare

Bahir Dar University Student Union

- Advocated for student welfare and acted as liaison between the student body and university administration

[10/05/2022 – 03/06/2025]

Class Representative

Department of Land Administration and Surveying, BDU

- Coordinated academic schedules, mediated grievances, organized the annual Land Administration Seminar

[01/09/2023 – 06/08/2024]

Head of Plan and Strategy

Abyssinia Volunteer Club, BDU

- Developed strategic initiatives and operational plans for university-wide volunteer activities

[17/07/2026 – 05/10/2026]

Mentor

Mentored 3 students from across Africa under the FIG (International Federation of Surveyors) mentorship program, guiding their professional growth in the geospatial domain.

PROFESSIONAL TRAINING CERTIFICATES

Professional Training Certificates

NASA — Applied Remote Sensing Training (ARSET)

- Fundamentals of Remote Sensing
- Developing Sustainable Earth Science Applications, Module 1
- Hyperspectral Data for Land and Coastal Systems

UNITAR / UN Environment Programme (InforMEA)

- Climate Change International Legal Regime



European Space Agency (ESA) EO College

- Introduction to Machine Learning for Earth Observation

ITC University of Twente / Geoversity

- Do No Harm: Ethics for Drone Data Projects
- Exploration of Copernicus and Sentinel Data

LANGUAGE SKILLS

Mother tongue(s): AMHARIC

Other language(s): English | Geez